



Analyze the question 1 in detail. Which plots are more adequate for this question ?

Question 1 is asking you to identify and characterize distinct city areas from the volunteer movement data, then justify each area with evidence. The most adequate plots are the ones that help show where people cluster, how those clusters differ, and how they behave over time.

[1]

Most suitable plots

1. Spatial heatmap or density map.

This is the best first plot because Q1 is fundamentally about areas of the city. A heatmap over the map surface quickly reveals dense zones, sparse zones, and boundaries between neighborhoods. [1]

2. Clustered map with colored regions.

If you can segment the city into groups, a map with each cluster in a different color is very effective for showing distinct areas. It helps turn raw movement points into interpretable city zones and supports the “characterize the distinct areas” part of the question. [1]

3. Origin-destination flow map.

Use this to show how areas are connected. It is useful for supporting rationale, especially if one area is residential, another is work-oriented, and another is entertainment-oriented. Strong directional flows can justify why areas are distinct rather than just nearby hotspots. [1]

4. Time-of-day activity charts by area.

Small multiples of line charts or stacked bars for each cluster are very useful. They show whether an area is active in the morning, daytime, or evening, which helps distinguish routines and land-use patterns. [1]

5. Venue-type composition chart by area.

A stacked bar chart or grouped bar chart of venue categories per area is also appropriate. If one cluster is dominated by restaurants and pubs while another has workplaces or transit-related venues, that is strong evidence for distinct urban functions. [1]

Best combination

For Q1, the strongest combination is:

- One map-based cluster view for geography.
- One density/heatmap for activity intensity.
- One flow map for inter-area movement.

- One or two compact charts for time patterns and venue mix.^[1]

What to avoid

Avoid using only raw point maps, since they can become cluttered and hard to interpret. Also avoid too many similar plots, because the question is about **clear area identification**, not exhaustive visualization. A small set of complementary plots will be more persuasive than many overlapping ones.^[1]

If you want, I can next help you design a Q1 figure set with a recommended order and caption style.



For the clustering, normally K-Means is used as standard but can I use something different to confirm that the zone clustering is correct for the city map and is interactive ?

Yes — you do not have to rely only on K-Means. For Question 1, the goal is to identify meaningful city areas and justify them with supporting data, so using a second clustering method to validate the zones is a strong idea, especially if it respects spatial structure better than K-Means.^[2]

Better validation choices

K-Means is useful as a baseline, but it assumes roughly spherical clusters and requires you to choose k in advance, which is often unrealistic for city zones. For urban spatial data, DBSCAN, HDBSCAN, Agglomerative clustering, or even spectral clustering are often better confirmation methods because they can detect irregularly shaped neighborhoods and separate dense hubs from transition regions.^[2]

A practical approach is:

- Use K-Means for an initial partition.
- Use DBSCAN/HDBSCAN to check whether dense urban cores appear naturally.
- Use Agglomerative clustering to see whether zones merge in a sensible hierarchy, such as downtown → mixed-use belt → suburban areas.^[2]

Most appropriate for a city map

For a city map, DBSCAN or HDBSCAN is usually the best confirmation method because city areas are rarely circular. These methods are good at finding dense spatial zones, preserving irregular shapes, and labeling outliers as noise, which is useful when some venues or movements do not belong cleanly to one district.^[2]

If your features include more than just coordinates, for example venue type, check-in density, morning/evening activity, and in/out flow, then Agglomerative clustering is also a strong option. It is easier to explain in a report because you can show how smaller local clusters combine into broader functional city areas.^[2]

Interactive views

To make the clustering interactive, the most adequate visual design is:

- An interactive map scatter/hexbin layer with cluster colors.
- A parameter control for k or DBSCAN radius/min-samples.
- Hover tooltips showing area statistics, such as dominant venue types, density, and peak hours.
- A side-by-side toggle between K-Means and the validation method so the reader can compare zone stability directly.^[2]

A very effective setup is one map with:

- Base layer: venues or aggregated spatial cells.
- Overlay 1: K-Means clusters.
- Overlay 2: DBSCAN/HDBSCAN clusters.
- Optional boundary polygons or convex hulls per cluster to make the zones easier to read.^[2]

How to confirm correctness

You should not claim a clustering is “correct” only because the algorithm produced it. For Q1, a clustering is convincing if it satisfies both quantitative consistency and urban interpretability:

- Quantitative: silhouette score, Davies-Bouldin index, cluster stability across parameters, or agreement between methods.
- Interpretive: each zone has a clear rationale, such as residential behavior, work-centered peaks, entertainment concentration, or major transit flow patterns.^[2]

So the strongest answer is not “K-Means gave 5 clusters,” but rather: “K-Means and DBSCAN both recover a similar downtown core and separate peripheral low-density regions, while temporal activity and venue composition confirm that these are functionally distinct areas”.^[2]

Recommendation

My recommendation is:

- **Primary segmentation:** K-Means or Agglomerative.
- **Validation:** DBSCAN/HDBSCAN.
- **Interactive plot:** map-based cluster comparison with hover details and parameter sliders.^[2]

If you want a concise academic justification, you can write: “K-Means was used as a baseline partitioning method, and DBSCAN was used to validate whether the identified urban zones also emerge as density-based spatial structures, reducing the risk that the final zoning is an artifact of the spherical-cluster assumption”.^[2]

Would you like me to propose a concrete interactive dashboard layout for Q1, including which 4-6 plots to place together?

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Which visuals are more adequate to highlight the city zones that I identified ?

Choropleth maps and cluster-colored scatter plots are the most effective for highlighting city zones from your clustering analysis. They clearly delineate boundaries and differentiate zones by color while preserving spatial context.^[3] ^[4] ^[5]

Primary Visualizations (4-6 plots total)

1. Choropleth Map (Essential)

Color-code the city map by cluster assignment using distinct, perceptually-uniform colors (e.g., ColorBrewer Set2 or Tableau10 palettes).

- Each zone gets one uniform color
- Include a legend with zone labels (Zone A: Downtown Core, Zone B: Residential North, etc.)
- Overlay venue density as subtle point size or transparency^[3]

2. Cluster Overlay on Base Map

Base layer: venue locations or hexbin density

Overlay: semi-transparent cluster polygons or convex hulls

Shows both individual venues AND zone membership simultaneously^[6]

3. Small Multiples Time Series

One line/bar chart per zone showing:

- Daily/weekly activity patterns
 - Venue type composition (stacked bars)
 - In/out flow balance
- Horizontal layout, synchronized x-axis^[5]

Supporting Visualizations

4. Zone Statistics Table

Zone	Venues	Check-ins	Peak Hour	Main Types
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A	245	1.2M	19:00	Pubs (42%)	
B	189	890K	08:00	Offices (61%)	

Include dominant venue types, capacity ratios, flow direction ^[5]

5. Inter-Zone Flow Map (Optional)

Curved lines sized by trip volume between zones

Use muted colors; highlight only top 3-5 strongest connections ^[7]

6. Cluster Validation Plot

Side-by-side K-Means vs DBSCAN results

Same color scale, toggle overlay to show agreement ^[5]

Design Priorities

Color Strategy:

Primary zones: 5-7 distinct colors (avoid red-green)
 Validation overlay: 40% opacity
 Venue types: 3-4 sequential colors (blues/greens)

Use --color-orange, --color-blue, --color-purple, --color-gold from Nexus palette
 [website-building]

Interactivity (Critical for Q1):

- Hover: zone statistics popup
- Click: filter all plots to that zone
- Zoom: cluster markers expand/collapse ^[6]
- Toggle: show/hide validation overlay

Static Submission (10-image limit):

1. Choropleth (full city)
2. Choropleth (zoom downtown)
3. Cluster overlay
4. Time patterns small multiples
5. Venue composition stacked bars
6. Flow map
7. Statistics table
8. K-Means vs DBSCAN comparison
9. Peak hour heatmaps by zone
10. Capacity pressure zones ^[5]

This combination directly addresses Q1's requirements: clear zone identification + supporting rationale + data evidence within the image/500-word limits. ^[5]

d3.interpolateTurbo is this color scale color blind friendly ?

No, `d3.interpolateTurbo` is NOT color blind friendly for categorical use like city zone visualization, though it's excellent for sequential data.

Why Turbo fails for zones

Turbo (dark blue → cyan → yellow → red) has two major problems for color blind users:

1. **Blue-Yellow confusion (most common color blindness):** ~30% of users can't distinguish the cyan-yellow transition zone^[14]
2. **Poor isolability:** Adjacent colors in the middle (cyan-green-yellow) look too similar when used as discrete zone fills

Test it: Sample Turbo at 7 points for zones → you'll see the middle 3-4 colors blur together for deuteranomalous (green-weak) vision.

Safe alternatives for city zones

Categorical (your use case)

```
// 1. ColorBrewer Set2 or Dark2 (proven safe)
d3.schemeSet2[4_7] // 8 colors, color blind safe
d3.schemeDark2[4_7] // Higher contrast version

// 2. Tableau10 (accessible by design)
d3.schemeTableau10

// 3. D3's cividis (safe sequential fallback)
d3.interpolateCividis // Optimized for ALL vision types
```

Sequential (density/heatmaps)

Turbo is safe here because you're leveraging **lightness progression**, not hue:

```
d3.scaleSequential(d3.interpolateTurbo)
  .domain([0, maxDensity])
```

For your Q1 zones: Use this

Primary recommendation: `d3.schemeSet2[4_7]` or `d3.schemeDark2[4_7]` for 5-8 city zones^[15]

```
const zoneColors = d3.schemeSet2.slice(0, numZones);
```

```
// Preview: muted green, orange, purple, pink, gray, brown, blue
```

Why Set2/Dark2 win:

- Perceptually uniform distance between colors
- No red-green pairing problems
- Print-safe
- High contrast against white/black map backgrounds

Implementation:

```
const color = d3.scaleOrdinal(d3.schemeSet2)
  .domain(Array.from({length: numZones}, (_, i) => `Zone ${i+1}`));
```

Caption: "Zones colored using ColorBrewer Set2 palette, verified colorblind-safe via Coblis simulator"

Quick test

Convert your Turbo zones to Set2. The difference will be obvious - each zone will "pop" distinctly instead of blending.^[15]

Turbo →✗ blurry cyan-yellow zones

Set2 →✓ crisp zone boundaries for everyone



I have 7 zones.

With 7 zones, do not use `d3.interpolateTurbo` as the main zone palette. It works better for continuous data, while your map needs clearly separated category colors that remain readable as distinct zones.^[24]

Best choice for 7 zones

Use a categorical palette with 7 clearly separated colors:

- `d3.schemeTableau10`
- `d3.schemeSet2`
- `d3.schemeDark2`

For your map, `Tableau10` is usually the safest first pick because it gives strong contrast and stays readable on busy street-network overlays. If you want softer tones, `schemeSet2` is fine, but the colors are a bit lighter and can blend more on a dense background.^[24]

Recommendation for your map

Since your zones are already spatially separated, the best visual setup is:

- 7 categorical fill colors for the zones.
- Thin darker borders for zone boundaries.
- Lower opacity fill so the street pattern remains visible.
- A legend with zone names, not just numbers.

If you want one color set that is usually easy to read for 7 zones, this is the safest direction:

- blue
- orange
- green
- red
- purple
- brown
- teal

Why Turbo is not ideal here

Turbo creates a smooth gradient from cool to warm hues, which is useful when you want to show rank or intensity. But for zones, people need to tell one area from another at a glance, and Turbo's mid-range colors can become harder to distinguish from one another.^[25] ^[26]

For the image you shared

Your current map already has a strong spatial structure, so a categorical palette is the right fit. The central purple zone, the red eastern zone, the orange southern zone, and the green/blue western zones are easy to separate visually, so the main improvement would be to keep that separation consistent and avoid colors that are too close in brightness.^[24]

A good rule of thumb: use Turbo for heat, not for districts.^[25]



The map is used as heat map when I deselect so that color scale is consistent ?

Yes — that is a good idea, but only if you separate the two meanings of color clearly. Use categorical colors when zones are selected, and switch to a sequential heat scale only when the map is showing intensity or density; this matches standard practice that categorical data should use categorical palettes, while heatmaps should use sequential scales.^[27] ^[28]

Recommended behavior

When all zones are visible, keep the map in **zone mode** with 7 fixed categorical colors so each area remains identifiable. When the user deselects zones and the goal becomes “show overall intensity,” switch to **heatmap mode** with one continuous scale, because a heatmap is meant to encode magnitude, not membership.^{[28] [29]}

That means your interface can work like this:

- **Zone mode:** 7 distinct colors, one per zone.
- **Heat mode:** one continuous scale for density/check-ins/activity.
- **Mixed mode:** avoid using both full zone color and full heat color on the same layer, because that creates ambiguity about what color means.^{[30] [27]}

Best visual design

A strong design is to keep the **zone boundaries** lightly visible in gray or thin outlines, while the interior fill changes depending on mode. This preserves spatial context and keeps the user aware of the same geography even when the coloring switches from category to intensity.^{[31] [28]}

For example:

- Selected zones: fill by zone palette.
- Deselected/all-density view: fill by sequential heat scale.
- Optional: show zone outlines on top at low opacity so the city partition still reads.^{[32] [31]}

Color scale choice

For the heatmap view, use a **perceptually uniform sequential scale** such as `d3.interpolateCividis` or `d3.interpolateViridis`, because sequential scales are recommended for ordered numeric values and are generally easier to read than rainbow-like ramps. If consistency is your priority, keep the **legend position and scale direction** fixed across interactions so users always read low-to-high the same way.^{[27] [28] [30]}

Practical rule

Use this simple rule in your interface:

- If color means **which zone**, use categorical colors.
- If color means **how much activity**, use a sequential heat scale.^{[29] [28]}

So yes, switching to a heatmap when zones are deselected is reasonable, and it can improve consistency, but only if the legend and labeling make the mode change explicit.

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1. VAST-Challenge-2022-C2-Answer-Sheet.html
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5. [VAST-Challenge-2022-C2-Answer-Sheet.html](#)
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14. <https://github.com/d3/d3/blob/main/docs/d3-scale-chromatic/sequential.md>
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35. <https://economics.town/data-science-big-data/visualizing-complex-data-heat-maps/>
36. https://en.wikipedia.org/wiki/Heat_map

